

The background image is a dark, blue-tinted photograph of a lush, green urban landscape. In the foreground and middle ground, there are dense clusters of trees and foliage. In the background, several tall, tree-like structures with wide, flat canopies (Supertrees) are visible against a cloudy sky. To the left, a large, dome-shaped building with a ribbed structure is partially visible. The overall scene suggests a sustainable, eco-friendly city environment.

WELCOME TO CDE2501 Liveable Cities

**SEMINAR 3 – SUSTAINABLE ENVIRONMENT
21 FEBRUARY 2025**

Lineup for Seminar 2

10.00am – 10.10am	Topic Introduction: “Ensuring a Sustainable Environment”
10.10am – 10.30am	Sharing by Ms Liu Huei Lyn , Assistant Director, Catchment & Waterways (Drainage Planning), PUB
10.30am – 10.50am	Sharing by Assoc. Prof. Adrian Loo , Deputy Director of the Centre for Nature-based Climate Solutions, NUS
10.50am – 11.00am	Synthesis
11.00am – 11.35am	Panel Discussion [Q&A]
11.35am – 11.40am	Closing & Admin Reminders to students

Ensuring a Sustainable Environment

What is ‘sustainability’?

1. ‘*Nachhaltigkeit*’ - comes from the field of forestry in Germany (Hans Carl von Carlowitz, 1713) – “**never harvest more than what the forest yields in new growth**”
2. Being popularised since 1987 by the Brundtland Report *Our Common Future* on sustainable development – “**development that meets the needs of the present without compromising the ability of future generations to meet their own needs**”
3. In today’s context, sustainability is about a **reconciliation** of **economic** growth and a solution to **social** and **environmental/ ecological** challenges (Kuhlman and Farrington, 2010).

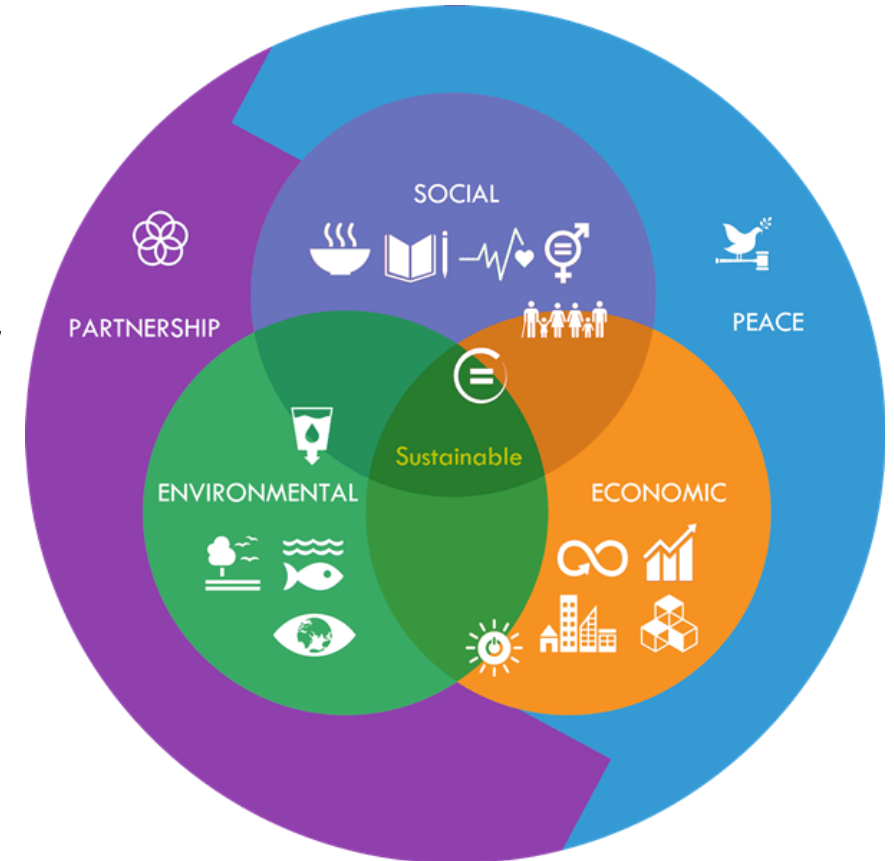


Image: 17 UN SDGs embedded in the three pillars of sustainability

From a scale of 1 to 5, how SUSTAINABLE is Singapore?

1 = very NOT sustainable

5 = very sustainable



[Pollev.com/liveablecities](https://pollev.com/liveablecities)

From a scale of 1 to 5, how sustainable is Singapore?

0

Very NOT sustainable — 1

2

3

4

Very sustainable — 5

From a scale of 1 to 5, how sustainable is Singapore?

0



From a scale of 1 to 5, how sustainable is Singapore?

0



SG Green Plan 2030

2024

GREEN PLAN

The Singapore Green Plan 2030 is our national sustainability movement, positioning us to achieve our target of net zero emissions by 2050. It is a living plan which continues to evolve. Here are the key updates and initiatives announced by our Green Plan Ministries during the Committee of Supply 2024.

[GreenPlan.gov.sg](https://www.greenplan.gov.sg)

JOINTLY LED BY

CITY IN NATURE

CONSERVING & EXTENDING OUR NATURAL CAPITAL

- Continue transforming Singapore into a City in Nature by conserving and extending our natural capital island-wide, such as:
 - Completed around 210km of Nature Ways and over 380km of park connectors
 - Planted more than 640,000 trees with the help of over 100,000 members of the community under our OneMillionTrees movement

SUSTAINABLE LIVING

BUILDING A CLIMATE-FRIENDLY SINGAPORE

- Enhance Climate Friendly Households Programme to encourage all HDB households to switch to energy and water efficient household appliances
- Expand energy labelling scheme for household water heaters and commercial storage refrigerators to help households and businesses make informed energy efficient choices
- Commence study to explore feasibility of expanding the Extended Producer Responsibility approach to cover more types of packaging waste beyond beverage containers

ENHANCING FAST CHARGING NETWORK

- Deploy more fast chargers to support the charging needs of fleet Electric Vehicle drivers

PILOTING "ADOPT A BLOCK" INITIATIVE

- Trial an "adopt-a-block" initiative where polytechnic and ITE students will promote sustainability, recycling and greening efforts, to residents living around their campus

ADVANCING ECO STEWARDSHIP PROGRAMME

- Deepen understanding of food sustainability for students through the Eco Stewardship Programme which was launched in 2021 for students to learn about Singapore's sustainability issues and actionable steps

ENERGY RESET

LAUNCH OF FUTURE ENERGY FUND

- Catalyse infrastructure investments for our energy transition

LAUNCH OF SUSTAINABLE AIR HUB BLUEPRINT

- Achieve net zero domestic and international aviation carbon emissions by 2050 through 12 initiatives across airport, airline and air traffic management domains. This includes:
 - Requiring flights departing Singapore to use Sustainable Aviation Fuel (SAF) from 2026. Targeting 1% SAF blend from 2026, to 3 - 5% SAF blend by 2030, subject to global developments
 - Maximising efforts to reduce energy use and deploy renewables, to reduce domestic aviation emissions from airport operations by 20% from 2019 levels in 2030

GREENING OUR BUILDINGS

- Introduce Mandatory Energy Improvement (MEI) regime which targets large, energy-intensive buildings by end 2024

ADOPTING BEST-IN-CLASS POWER GENERATION TECHNOLOGY

- Awarded YTL PowerSeraya the right to build, own and operate a new hydrogen-ready Combined-Cycle Gas Turbine (CCGT) power plant
- Commencement of operations by end 2027, making it the third hydrogen-ready CCGT power plant in Singapore

ADVANCING EFFORTS IN EXPLORING LOW-CARBON ALTERNATIVES

- Launched request for proposals for low/zero carbon ammonia solutions for power generation and bunkering
- Committed \$53 million to 16 Low-Carbon Energy Research projects to improve technical and commercial viability of low-carbon energy technologies such as hydrogen

GREEN ECONOMY

BUILDING SUSTAINABLE ENTERPRISES

- Introduced a new course to train up Zero Waste Managers to champion 3R (Reduce, Reuse, Recycle) efforts in their organisations

TRANSFORMING & DECARBONISING SECTORS

- Enhance Resource Efficiency Grant for Emissions to support more industrial facilities and projects to improve energy efficiency and reduce emissions

SEIZING OPPORTUNITIES IN THE LOW-CARBON ECONOMY

Helping businesses

- Extend Enterprise Financing Scheme-Green and expand the scheme to support adoption of green solutions
- Provide funding support for businesses starting on their sustainability reporting journey

Supporting workforce to upskill & reskill

- Develop training programmes in Carbon Management, Services and Trading with Institutes of Higher Learning

DEVELOPING CARBON CAPTURE & STORAGE AS A DECARBONISATION PATHWAY

- Work with an industry consortium to study the viability of developing a cross-border Carbon Storage project capturing emissions from Singapore

SUPPORTING BUSINESSES TO PURSUE SUSTAINABILITY COLLABORATIONS

- Introduce Environmental Sustainability Business Collaboration Guidance Note

ENCOURAGING BUSINESSES TO INVEST IN ENERGY EFFICIENT EQUIPMENT

- Expand Energy Efficiency Grant to support adoption of energy efficient equipment in more sectors

SUPPORTING MARITIME BUSINESSES TO TAP ON NEW GROWTH OPPORTUNITIES

- Establish a new pillar under the Maritime Cluster Fund to catalyse first mover adoption of pre-approved sustainability solutions in targeted new areas

RESILIENT FUTURE

SECURING A SUPPLY OF SAFE FOOD

- Better protect consumers and safeguard food supply resilience through upcoming Food Safety and Security Bill
- Accredited new food import sources in 2023 (e.g. Indonesia for live chickens and eggs, Türkiye for eggs)
- Piloted a Food Resilience Preparedness Project involving secondary school students to educate them on the importance of food security
- Transform our aquaculture sector through the Singapore Aquaculture Plan, which will embody our collective vision for productive and sustainable aquaculture
- Partner stakeholders to better match supply and demand for local produce via an industry-led aggregator, as well as develop a uniquely Singaporean fish product
- Onboard more food establishments to the Farm-To-Table Recognition Programme

STRENGTHENING CLIMATE RESILIENCE

- Propose new legislation and Code of Practice to support coastal protection measures
- Conduct site-specific studies at City-East Coast, North-West Coast and Jurong Island to better understand Singapore's coastlines
- Continue to engage stakeholders on coastal protection efforts including for Long Island

GREEN GOVERNMENT

PRESSING ON TOWARDS NET ZERO: GREENGOV.SG

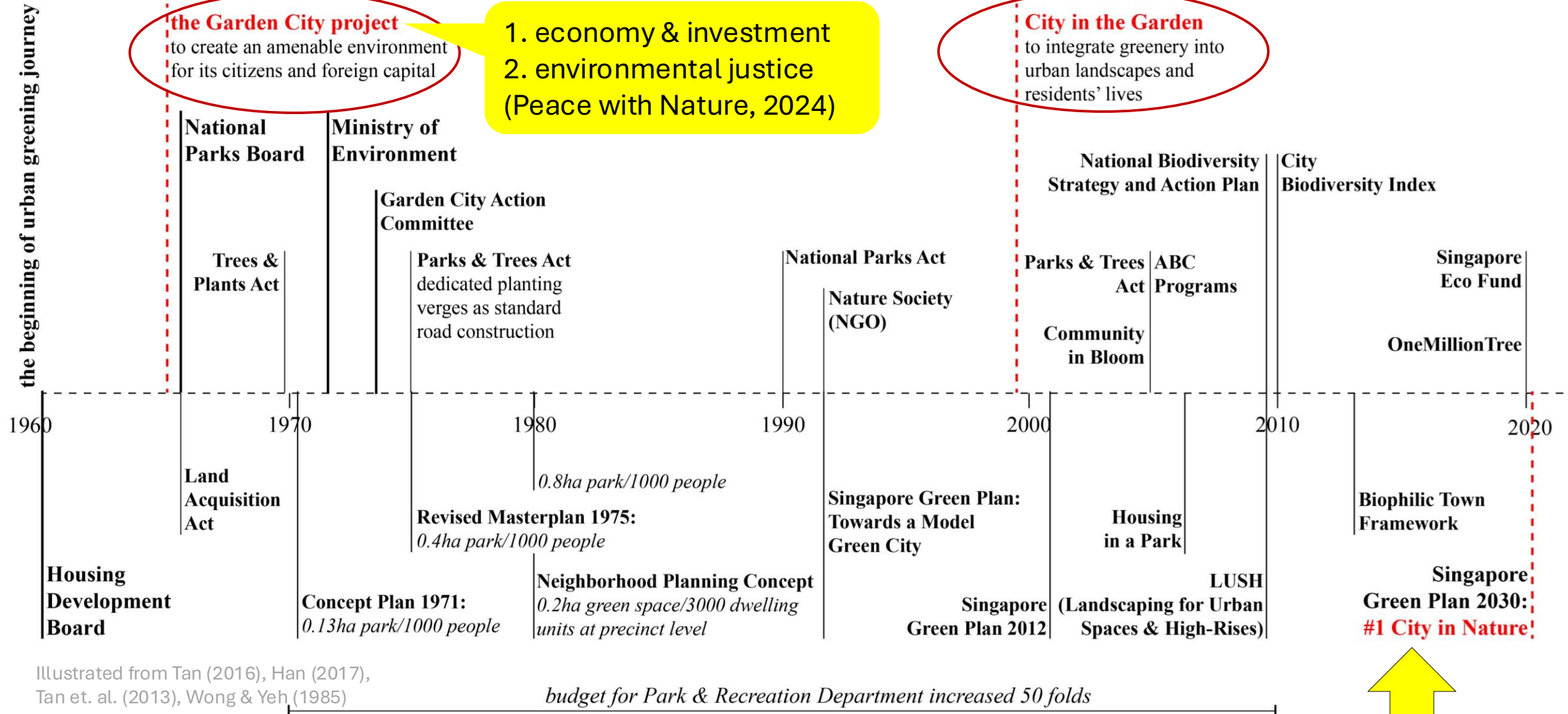
- Expand scope of tender evaluation points for sustainability to MICE events
- Inclusion of waste data in future GreenGov.SG reports

This scene stretches from Pearl's Hill on the left to Tanjong Rhu on the right. In the centre is Government Hill (now Fort Canning).



Image: National Museum of Singapore,
National Heritage Board

From a Garden City to a City in Nature



Illustrated from Tan (2016), Han (2017),
Tan et. al. (2013), Wong & Yeh (1985)



From a Garden City to a City in Nature

“I have always believed that a blighted urban landscape, a concrete jungle destroys the human spirit. We need the greenery of nature to lift up our spirits”

Lee Kuan Yew, the opening of National Orchid Garden, 20 October 1995

Singapore’s vision of a **City in Nature**:

“where greenery and nature are **harmoniously woven into the built environment** to support biodiversity conservation and so that residents can enjoy the benefits of a **healthy and biodiverse urban ecosystem**”

CLC, 2024 in “Building Liveable and Sustainable Cities – A Framework for the Future”

How to ensure a sustainable environment through the lens of a City in Nature?

Ms Liu Huei Lyn
Assistant Director
Catchment & Waterways (Drainage
Planning), PUB



Assoc. Prof. Adrian Loo
Deputy Director of the Centre for
Nature-based Climate Solutions,
NUS



Sustainable Environment through ‘City in Nature’ lens

Several key takeaways (there are more!):

1. An interplay of quantity and quality through integrated and multifunctional interventions

- a) Projects such as Bishan-AMK and Alkaff Lake showcase that **land constraint is not the limit**.
- b) There are state-of-the-art creativities in **intertwining engineering solutions, biodiversity conservation, and people-centric design** (including a sense of place, affinity towards nature).
- c) Such a close-knit interplay requires **interdisciplinary knowledge and collaborations**.



Sustainable Environment through ‘City in Nature’ lens

Several key takeaways (there are more!):

2. Consistent and long-term commitment is a key enabler

- Commitment from the government (including the community) ensures the **continuing delivery** of the roles of nature in fostering a sustainable environment.
- Such commitment is also by **well-informed evidence** of the roles of nature in fostering a sustainable environment (e.g. biodiversity, health and well-being, economic competitiveness)

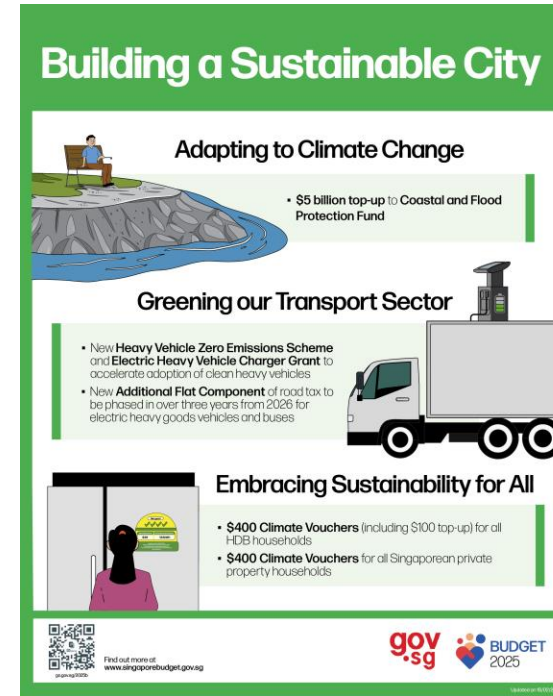


Image: Singapore Budget 2025, MOF

e.g., How can nature be integrated with climate change adaptation measures?

Budget debate: \$315m to expand and enhance parks, park connector network and recreational routes



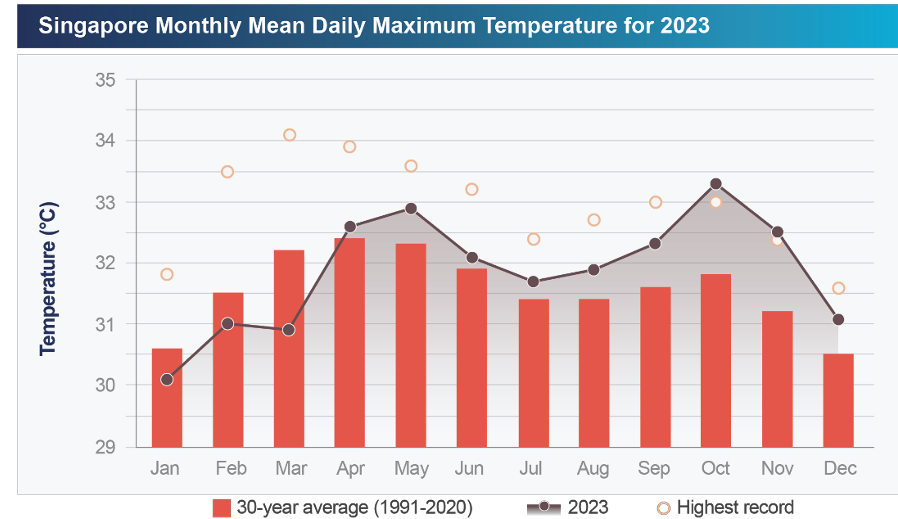
Artist impression of a new 7.6ha park to be built in Teachers' Estate near Upper Thomson by 2024. PHOTO: NATIONAL PARKS BOARD

Sustainable Environment through ‘City in Nature’ lens

Several key takeaways (there are more!):

3. Long-term perspectives

- Interplay between sustainability and resilience in dealing with the **new reality** and **uncertainty** of environmental and societal challenges.
- Appreciation of **varying preferences, perceptions and needs** towards nature and the environment.
- Thinking **beyond our shores** – transboundary impacts of our sustainability initiatives and how others also impact us



Climate change-fuelled heat stress could cut productivity and fertility, costing billions: NUS study



BY DEBORAH LAU
Published March 18, 2024
Updated March 19, 2024

Source:
<https://www.todayonline.com/singapore/climate-change-heat-stress-cut-productivity-fertility-cost-billions-nus-study-2385046>

Sustainable Environment Beyond Blue and Green

Sustainable Environment also emphasises:

1. **resource security** : “*environmental protection and responsible use of resources*” → the initial concept of sustainability!
2. **energy security** – *trilemma*: environmental sustainability, economic competitiveness, energy security → **trade-offs/ harmony**
3. **circular resource** usage e.g., recycled concrete, repurposing of buildings and waste
4. building climate and social **resilience**

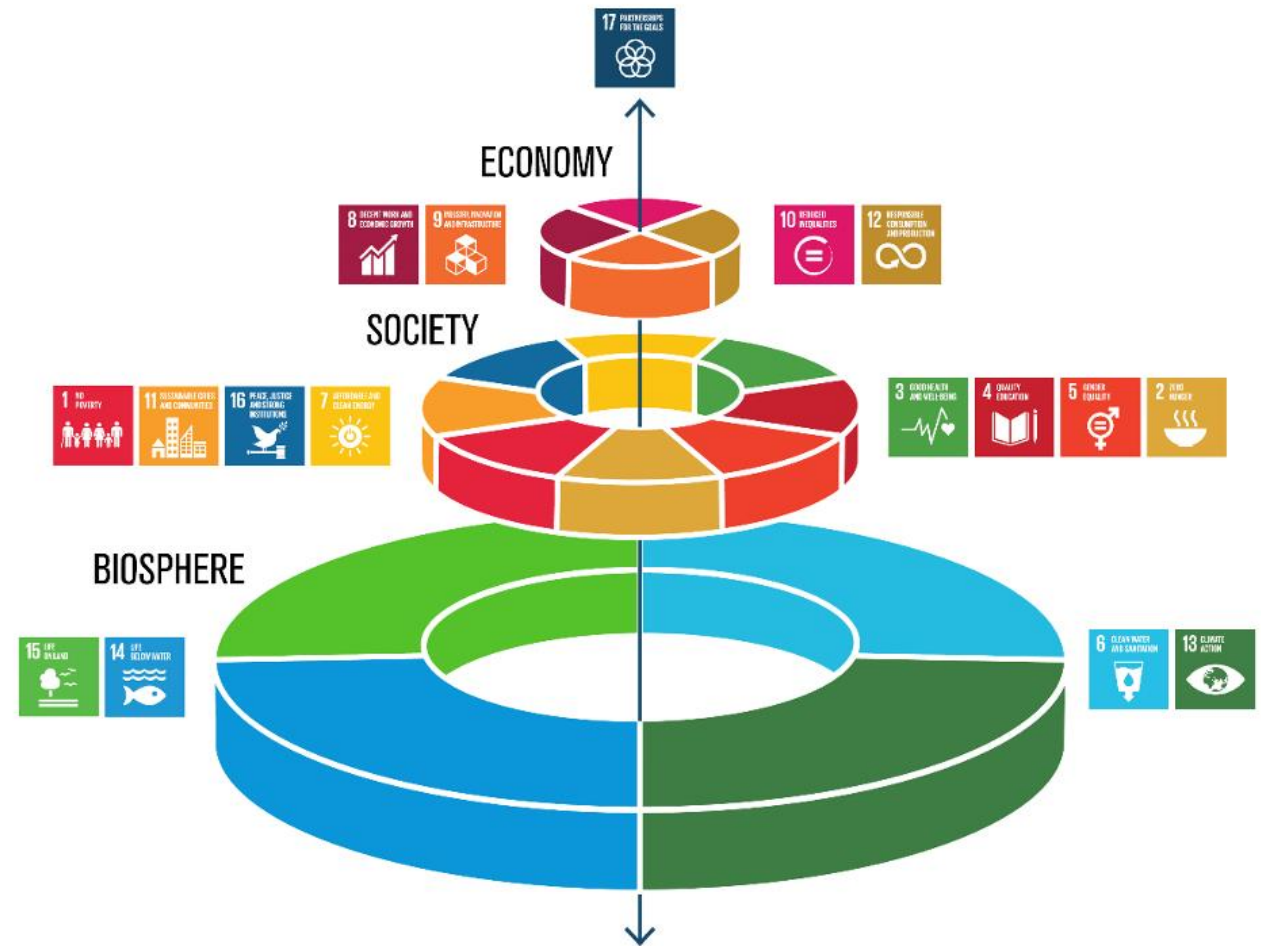


Image: UN 17 SDGs, World Sustainability Collectives

**How do we
ensure a sustainable environment**



a high quality of life ?

housing needs VS nature preservation ?

Dover Forest to be used for both housing and nature; first housing project to be launched in 2022



This decision was made after feedback from members of the public and the findings of two separate scientific studies. ST PHOTO: LIM YAOHUI

how to live with 'nature'?

30 monkeys cause chaos in Punggol on 2nd day of CNY, break into residents' homes & steal food



LATEST NEWS SINGAPORE

Some monkeys climbed along balconies and entered lower-floor units to steal food.

By Gary Yang - 2 Feb 2025, 12:03 pm

Source: <https://mustsharenews.com/monkeys-punggol-chaos/>

Expectations for final presentation

Show the link between **High Quality of Life** and **Sustainable Environment** and/or **Competitive Economy** (*Week 7 Seminar*):

1. **Refine** your **Problem Statement** by **expanding** your High Quality of Life to include **one or both** of these outcomes
2. **Explain** how the problem is related to sustainability/ environment **and/or** the economy in your problem analysis

Final presentation deliverables

- **Brief Context of the site** (historical, planning/ development, socio-cultural contexts etc)
- **Identification of the problem(s)** for the given site, and summarising it through the **Problem Statement (refined from Presentation 1)**
- **Show the link between High Quality of Life and Sustainable Environment and/or Competitive Economy**
- **Analysis of the site and problem(s)** through tools like System's Map or Causal Loop Diagram
- **Describe the relevant stakeholders/communities** connected to the identified problem.
- **Solution(s)** addressing the root causes of the problem
- **Evaluation of the impact on the solution(s)** on the relevant stakeholder/communities and the possible trade-offs

Quiz #1: Sustainable Environment!

- Please refresh your Canvas page:
 - Navigation Bar > Assignments > Quizzes > Quiz 1: Sustainable Environment
- Quiz is open for 20 minutes!
- You have **8 minutes** to answer **5 multiple choice questions**.
- Open internet and open book, **NO collaboration**
- Password (uppercase): **H2O**

Q & A

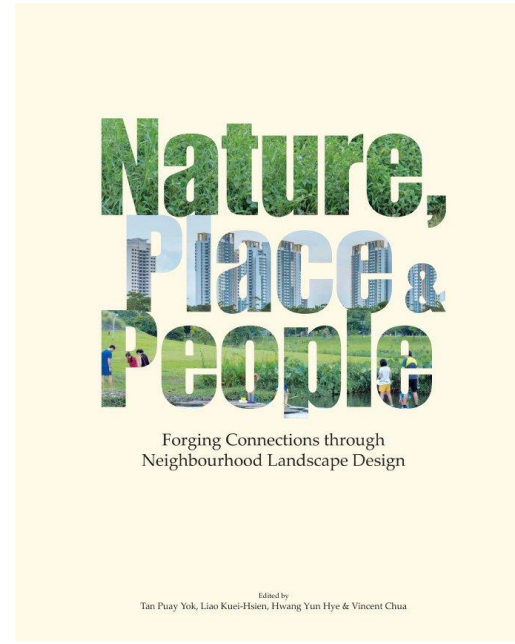
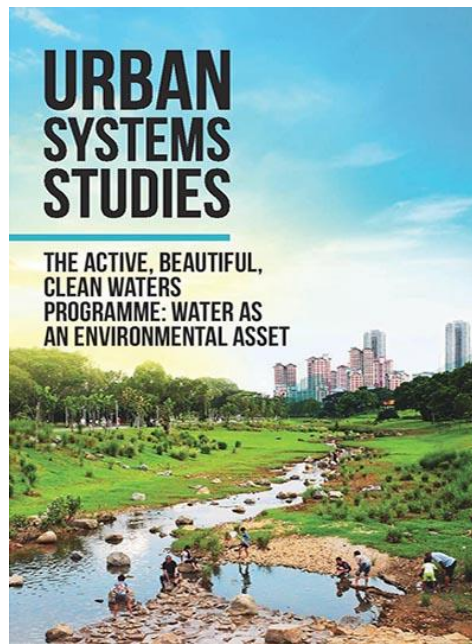


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Nobody has responded yet.

Hang tight! Responses are coming in.

Recommended Readings



Week 7, Friday 7 March, 10:00am to 12:00pm

Seminar 4: Creating a Competitive Economy

Speakers/Panelists:

- **Lim Chin Chong:** Former Director of Specialised Parks Development Group, JTC
- **Adj. Prof. Koh Choon Fah:** Adjunct Professor, NUS Cities; Former CEO of Edmund Tie & Company; Senior Accredited Board Director, Singapore Institute of Directors
- Moderated by **Professor H. Koon Wee:** Visiting Associate Professor, Department of Architecture, NUS

Topics covered:

- The role of real estate in a competitive economy
- The role of industrial parks in a competitive economy

Singapore and Vancouver | NUS-UBC Joint Summer Studio

The "Liveable City": A Comparative Exposition

"LIVING WITH WATER"

Singapore: 1 – 10 June 2025
Vancouver: 13 – 21 June 2025



E-mail announcement will be sent out soon!